

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building production LLM-powered RAG systems and agentic workflows, with a strong track record of end-to-end delivery: from understanding customer pain points to designing, building, and shipping scalable AI systems. Proven production metrics: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% fewer post-deployment defects. Python- and TypeScript-proficient with FastAPI, LangChain, LlamaIndex, Docker, and observability tooling - ready to build intelligent solutions for Ramp's largest enterprise customers.

CORE COMPETENCIES

Agentic LLM Systems • RAG Pipeline Design • Enterprise AI Deployment • Python & TypeScript • FastAPI / Flask • LangChain & LlamaIndex • Docker & CI/CD • Prompt Engineering • Observability & Logging • React & Full-Stack AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built **agentic LLM workflows** for multi-step healthcare queries with tool discipline and structured reasoning; improved agent response stability ~25% and reduced hallucinations >30% - same pattern needed for Ramp's enterprise AI layer.
- Designed and deployed **production RAG systems** (recursive semantic chunking, transformer embeddings, evaluation pipeline); improved contextual retrieval precision ~35% and reduced irrelevant retrieval >30%.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker, with structured logging and load simulation; reduced post-deployment defects ~30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for customer risk scoring; improved recall 15-20% on high-risk categories while holding precision >90%.
- Built preprocessing pipelines raising dataset reliability above 98% and cutting downstream model instability ~40%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP, Industrial Automation | Hyderabad, India

- Optimized SQL/T-SQL stored procedures powering the Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed Grafana performance dashboards tracking long-running queries and deadlocks; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Video Analytics, Applied NLP | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time 60-70% and achieved ~85% alignment with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30% and raised contextual accuracy >85%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated JSON contracts between agents, RAG-grounded validation, approximate-nearest-neighbor duplicate detection, and explainable risk scoring with audit traces. Demonstrates end-to-end agentic AI delivery on complex enterprise data.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, multi-agent orchestration

Dream Decoder - Multimodal AI Orchestration Platform

Full-stack FastAPI + React/TypeScript/Vite app coordinating multiple AI APIs; introduced intermediate structured prompt transformation layers - improved contextual alignment ~30% and first-pass success rate ~25-30% over direct-prompt orchestration.

FastAPI, React, TypeScript, Vite, Tailwind, Perplexity Sonar API, Replicate

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, [Chrome Web Store](#)

Integrates four AI vision providers; multi-section capture pipeline, 7-day translation cache, 29-site support. Demonstrates independent end-to-end product delivery with production AI integration.

TypeScript, Manifest V3, Claude Sonnet, GPT-4o mini, multimodal AI

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, SQL (T-SQL, PostgreSQL)

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Enterprise AI, Financial AI Automation, Healthcare AI, Workflow Automation